

Evolution of Medical LLM Training Systems

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Abstract

Medical large language models (LLMs) have rapidly evolved from instruction-tuned chatbots into increasingly autonomous training systems for clinical knowledge, reasoning, and self-improvement. However, the central systems question remains unresolved: how should a general LLM be transformed into a reliable medical expert without excessive data engineering, catastrophic forgetting, expensive human supervision, or over-reliance on static teacher demonstrations?

This thesis reviews and synthesizes three representative works from the HuatuoGPT series: HuatuoGPT-II, HuatuoGPT-o1, and HuatuoGPT-3. These works form a coherent progression in medical LLM training systems. HuatuoGPT-II studies how to inject large-scale medical knowledge efficiently by replacing the conventional continued-pretraining-plus-SFT pipeline with one-stage domain adaptation based on data unification and priority sampling. HuatuoGPT-o1 shifts the focus from knowledge acquisition to complex reasoning, introducing verifiable medical problems and a two-stage verifier-guided training framework that combines searched complex chain-of-thought supervision with reinforcement learning. HuatuoGPT-3 further questions whether supervised fine-tuning is necessary at all, proposing an RL-only adaptation paradigm with off-policy seeding, accelerated off-policy learning, and automatic seed phase-out.

The core argument of this thesis is that medical LLM training systems are evolving along a clear trajectory: from data-heavy knowledge injection, to

feedback-driven reasoning enhancement, and finally to autonomous policy optimization. Across the three works, the training objective becomes increasingly aligned with the deployment objective. The data pipeline becomes less dependent on manually curated demonstrations. The role of external models changes from permanent teachers to temporary bootstrap signals. The optimization loop also moves from static imitation toward self-evolution under verifiable and rubric-based rewards.

Through a unified comparative framework, this thesis analyzes each work’s motivation, method design, experimental evidence, system-level trade-offs, and limitations. It concludes that future medical LLM training systems should integrate three principles: unified data interfaces, reliable domain-specific feedback, and adaptive mechanisms that preserve exploration after absorbing external knowledge. Open challenges remain in clinical safety, reward design for open-ended medicine, multimodal grounding, continual updating, and rigorous deployment evaluation.

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Chapter 1

Introduction

1.1 Research Background

Large language models have become a general-purpose interface for knowledge-intensive work. In medicine, however, simply applying a general model is insufficient. Clinical tasks require specialized knowledge, careful reasoning, transparent uncertainty handling, and sensitivity to safety. A fluent but medically shallow response may be worse than no response, because users can mistake linguistic confidence for clinical reliability.

This motivates the development of medical LLMs: models adapted to understand medical concepts, answer domain-specific questions, reason over symptoms and evidence, and communicate in a clinically appropriate style. Early medical LLM work mainly focused on instruction tuning and domain-specific data collection. More recent systems increasingly emphasize reasoning, reward feedback, and autonomous improvement. The HuatuoGPT series is a useful lens for studying this evolution because its successive systems target different bottlenecks in the medical adaptation pipeline.

From a systems perspective, training a medical LLM is not a single algo-

rithmic choice. It is a pipeline that connects data collection, data transformation, sampling policy, objective design, feedback construction, evaluation, and deployment assumptions. Each design decision affects both model quality and engineering cost. A two-stage pipeline may look modular, but it introduces distribution shifts and duplicated hyperparameter tuning. A pure RL pipeline may look elegant, but it can fail when the model cannot discover domain knowledge by itself. A verifier-based reasoning system can reduce human labeling cost, but only if verification is reliable enough for medical semantics.

1.2 Core Problem

The central problem studied in this thesis is:

How can training systems transform general LLMs into capable medical LLMs effectively and efficiently?

This question is not only about model accuracy. It is about the design of a training system. The system must decide how medical data enter the model, how the objective changes during adaptation, how feedback is constructed, and when the model should imitate external demonstrations or explore beyond them. The word “effectively” refers to medical capability: the model should answer, reason, and communicate in ways that match domain requirements. The word “efficiently” refers to the training system: the pipeline should reduce avoidable data conversion, repeated training phases, unstable objective switching, and dependence on permanent teacher demonstrations. In this thesis, the three selected works correspond to three increasingly ambitious answers: first unify the knowledge-injection pipeline, then introduce verifiable reasoning feedback, and finally make reinforcement learning itself the main adaptation engine.

1.3 Three Core Papers

The logical progression of the three core papers is shown in Figure 1.1. They are not treated as isolated model releases. Instead, they are interpreted as a connected research line in which each work answers a limitation left by the previous one.

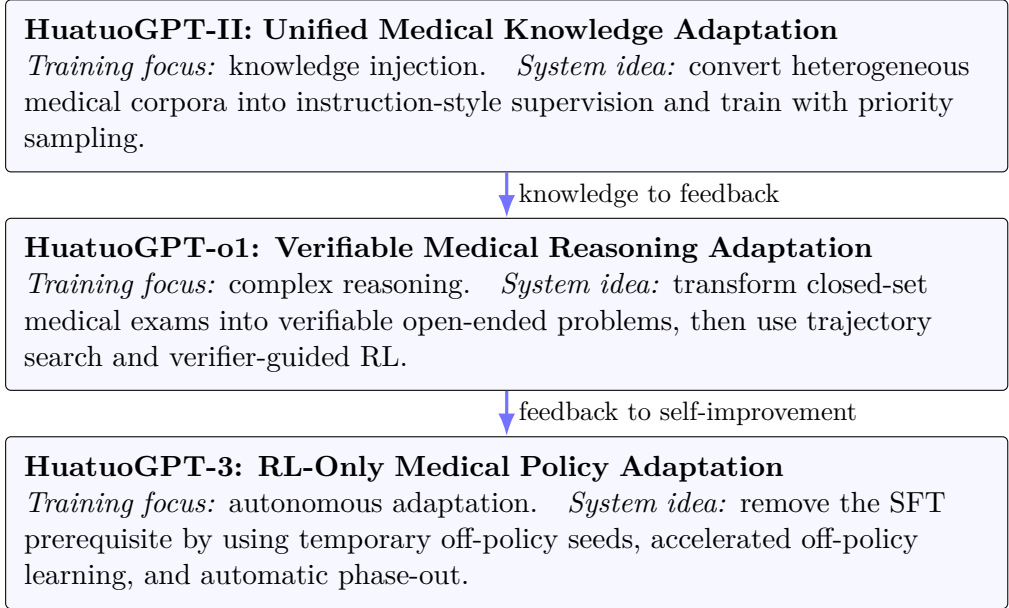


Figure 1.1: Progressive research trajectory of the three core papers: from unified knowledge adaptation to verifiable reasoning reinforcement and RL-only policy adaptation.

The three papers gradually move the training system from data formatting toward feedback and policy optimization. HuatuogPT-II asks how to make domain knowledge enter the model more smoothly. HuatuogPT-o1 asks how to make the model reason under medical verification. HuatuogPT-3 asks how to let the model adapt through RL without being fixed to supervised imitation.

The timeline is further summarized in Figure 1.2. This thesis focuses on the three training-system papers most directly related to the evolution from supervised adaptation to RL-based self-improvement.

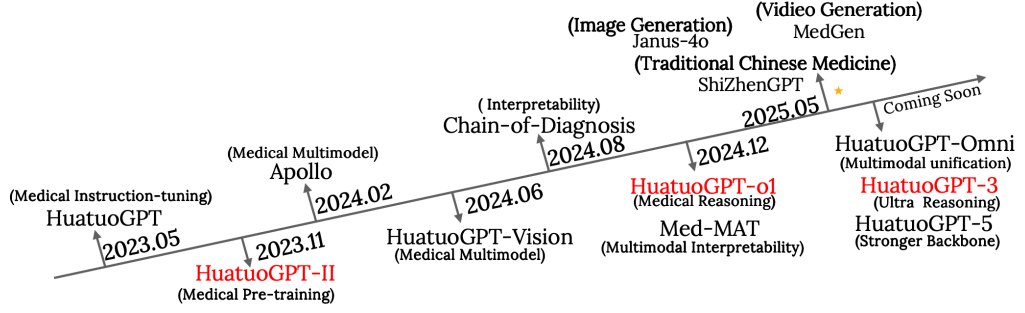


Figure 1.2: Timeline of the HuatuoGPT series. This thesis focuses on HuatuoGPT-II, HuatuoGPT-o1, and HuatuoGPT-3 as three connected training-system designs in the evolution of medical LLMs.

1.4 Thesis Contributions

The broader HuatuoGPT line has had visible influence in medical LLM research, including text-only, reasoning-oriented, multimodal, and RL-based systems. Table 1.1 summarizes representative versions and their academic impact. This context matters because the three papers studied in this thesis are not isolated prototypes; they are part of a sustained series that repeatedly targets new bottlenecks in medical LLM training.

Version	Key Feature	Stars	Model D/L	Data D/L	Venue	Cites
HuatuoGPT-1	Fine-tuned LLMs to behave as doctors	1,233	8,681	2,876	EMNLP 2023	481
HuatuoGPT-2	Injected large-scale medical knowledge into LLMs	387	13,093	2,740	COLM 2024	154
HuatuoGPT-Vision	Built multimodal medical LLMs with visual reasoning	318	354,047	11,370	EMNLP 2024	234
HuatuoGPT-o1	Introduced explicit medical reasoning capabilities	1,146	23,615	87,719	ACL 2025	305
HuatuoGPT-3	Proposed RL-only medical policy adaptation	—	—	—	ICML 2026	—

Table 1.1: Overview of the HuatuoGPT series: features, popularity, and academic impact. D/L denotes downloads.

The contribution of this thesis is a unified systems-level reconstruction of the three works. Rather than summarizing each paper independently, it explains how the training objective, data interface, feedback source, and teacher role change across the HuatuoGPT series. It also compares the three systems along common axes such as supervision source, reward design, optimization objective, engineering complexity, and failure mode. Finally, it identifies open challenges for future medical LLM systems, especially reward reliability, multimodal grounding, safety evaluation, and continual adaptation.

1.5 Thesis Organization

Chapter 2 introduces the background of medical LLMs and domain adaptation. Chapter 3 studies HuatuoGPT-II as unified medical knowledge adaptation. Chapter 4 studies HuatuoGPT-o1 as verifiable medical reasoning adaptation. Chapter 5 studies HuatuoGPT-3 as RL-only medical policy adaptation. Chapter 6 synthesizes the three systems and discusses open challenges. Chapter 7 concludes the thesis.

Chapter 2

Background and Related Work

2.1 Medical LLMs

Medical LLMs adapt general-purpose language models to healthcare tasks such as clinical question answering, medical exam solving, triage, patient communication, report generation, and decision support. General foundation models such as GPT-4 [1], LLaMA [44], and LLaMA-3 [15] provide broad linguistic and reasoning abilities, while medical systems specialize those abilities toward clinical use. Representative medical LLMs include Med-PaLM and Med-PaLM 2 [40, 41], GPT-4-based medical prompting studies [30], Meditron [10], PMC-LLaMA [50], MedAlpaca [17], ChatDoctor [22], DISC-MedLLM [3], BianQue [5], DoctorGLM [51], BioMistral [21], OpenBioLLM [34], UltraMedical [55], ApolloMoE [6], and HuatuoGPT [57]. These systems show that medical capability can be improved through domain-specific data and alignment, but they also reveal persistent limitations: hallucination, weak reasoning under ambiguity, poor calibration, and dependence on costly supervision.

The medical domain differs from many general NLP tasks in three ways. First, errors can have direct safety implications, which motivates medical eval-

uation protocols beyond ordinary language quality [2, 30, 40]. Second, factual correctness often depends on subtle clinical context rather than surface keyword matching, as reflected by exam-style and expert-level benchmarks such as MedQA, MedMCQA, Medbullets, and MedXpertQA [7, 19, 33, 59]. Third, useful answers require not only retrieving knowledge but also reasoning over symptoms, probabilities, contraindications, and patient-specific constraints [11, 41, 55]. A model that memorizes medical facts may still fail when it must compare differential diagnoses or decide which missing information should be requested next. Therefore, a training system for medical LLMs must address both knowledge and reasoning.

2.2 Domain Adaptation Pipelines

The standard adaptation route for specialized LLMs usually contains two stages. The first stage performs continued pretraining on domain corpora to inject terminology and factual knowledge. The second stage performs supervised fine-tuning (SFT) on instruction-response pairs to align the model with user-facing interaction formats. This route has been widely used in medical LLMs and other vertical-domain models [9, 10, 42, 50, 53, 55]. Similar specialization pipelines have also appeared in legal and financial domains, showing that domain adaptation is a general systems problem rather than a medicine-only problem [4, 24, 43, 58]. Recent studies further question whether continued pretraining is always necessary, because domain specialization can also be achieved through higher-quality instruction mixtures, careful curriculum design, and downstream objective alignment [8, 16, 29]. The pipeline is intuitive, but it has several systems-level weaknesses.

First, the two stages optimize different objectives. Continued pretraining

predicts raw text; SFT predicts task responses. When the objective changes abruptly, optimization can suffer from distribution shift, capability regression, or forgetting [8, 16, 29]. Second, the two stages require separate data curation and hyperparameter tuning, increasing engineering overhead [9, 42, 53]. Third, if the domain corpus is heterogeneous, noisy, multilingual, or written in styles unlike user instructions, continued pretraining may inject knowledge without teaching the model how to use it in dialogue [9, 32, 47]. This is why HuatuoGPT-II treats data format itself as a systems object rather than a preprocessing detail: a corpus is valuable only when the training objective can turn it into deployable behavior [9, 55, 57].

2.3 Reasoning and Reinforcement Learning

Recent LLM research has shown that reasoning can be improved by scaling chain-of-thought generation, verifier feedback, and reinforcement learning. Chain-of-thought prompting [49] demonstrates that explicit intermediate reasoning can improve multi-step problem solving. RLHF and preference optimization methods such as human-preference RL [13], InstructGPT [32], PPO [38], DPO [36], RLOO [20], and ReMax [23] show how behavior can be optimized beyond plain supervised learning. More recent reasoning systems use GRPO-style training [39], DAPO-style large-scale RL [54], and DeepSeek-R1-like self-improvement [14, 26]. These developments are closely related to medical reasoning because they move model training from passive imitation toward candidate generation, scoring, and revision [11, 12, 31]. In domains such as mathematics and code, reward construction is relatively direct: an answer can be checked by exact match, symbolic verification, or unit tests [14, 39]. Medicine is more difficult. Many medical answers have aliases, acceptable variants, or context-dependent formulations. In-

intermediate reasoning steps are also hard to verify automatically [2, 11, 41].

Nevertheless, medical exams provide a useful bridge. Multiple-choice benchmarks such as MedQA [19], MedMCQA [33], PubMedQA [18], CMB [46], CMExam [25], MMLU-Pro [45], and GPQA [37] contain objective answers, while their clinical stems still require domain reasoning. Newer medical evaluations such as Medbullets, MedXpertQA, and HealthBench further stress difficult exam knowledge and open-ended clinical response quality [2, 7, 59]. If such questions can be transformed into open-ended problems with unique answers, then a verifier can score the final answer and provide a reward signal [11]. This is the key insight behind HuatuoGPT-o1. It also explains why the paper’s verifier is not an auxiliary evaluation trick but a central training-system component.

2.4 Toward Autonomous Training Systems

SFT remains useful because it efficiently teaches a model to imitate high-quality demonstrations [32, 47]. However, imitation can also constrain exploration. If SFT collapses the policy toward a narrow distribution, later RL may spend many updates escaping the teacher’s behavior instead of discovering better reasoning paths [28, 29, 48]. Pure RL avoids this imitation bottleneck, but it may fail when the base model cannot sample useful domain behaviors on its own [14, 26]. Recent off-policy and mixed-policy RL methods attempt to combine demonstrations with exploration [27, 35, 52, 56], but persistent teacher trajectories can anchor the policy to the teacher distribution.

The main challenge is therefore to combine external guidance with autonomous exploration. HuatuoGPT-3 addresses this by treating teacher outputs as temporary seeds inside an RL-only process, building on the broader movement from static supervised adaptation toward on-policy reasoning opti-

mization [14, 39, 52, 54, 56]. The system uses off-policy trajectories early to bootstrap capability, but phases them out once the model’s on-policy outputs become stronger [12]. This converts the role of the teacher from a permanent imitation target into a transient accelerator. In this sense, the three HuatuoGPT systems are not merely three model generations; they are three increasingly autonomous training systems [9, 11, 12, 57].

Related-Work Theme	Observed Bottleneck	HuatuoGPT Response and Thesis Role
Domain adaptation	Continued pretraining and SFT optimize different objectives and require separate data engineering [8, 29, 42].	HuatuoGPT-II: unifies heterogeneous medical corpora as instruction pairs and studies whether one-stage adaptation can reduce objective mismatch.
Verifier-guided reasoning	Medical reasoning needs more than final-answer imitation, but dense expert labels are too expensive [2, 11, 49].	HuatuoGPT-o1: converts exam questions into verifiable open-ended tasks so reasoning trajectories can be searched, filtered, and improved by RL.
Autonomous RL	SFT can bootstrap behavior but may narrow exploration; pure RL explores policy seeds inside RL, then phases out when on-policy responses become stronger [14, 28, 52].	HuatuoGPT-3: uses temporary off-policy seeds inside RL, then phases them out when on-policy responses become stronger.

Table 2.1: Relationship between related-work bottlenecks and the three HuatuoGPT training-system responses.

Table 2.1 summarizes the connection between the related work and the three papers studied in this thesis. The table is placed here because the three systems should be read as answers to known training-system bottlenecks: HuatuoGPT-II addresses data and objective mismatch, HuatuoGPT-o1 addresses scalable reasoning feedback, and HuatuoGPT-3 addresses the tension between external guidance and autonomous exploration.

Chapter 3

HuatuoGPT-II: Unified Medical Knowledge Adaptation

3.1 Motivation

HuatuoGPT-II addresses the first bottleneck in medical LLM development: how to inject large-scale medical knowledge into a general LLM without paying the full cost of a brittle two-stage pipeline. The conventional recipe is continued pretraining followed by SFT. In medicine, this recipe is especially cumbersome because the domain corpus may include textbooks, encyclopedias, web documents, academic literature, and multilingual sources. These data types differ sharply from user-facing medical instructions, both in language style and in the granularity of supervision.

The key observation of HuatuoGPT-II is that the distinction between “pre-training corpus” and “fine-tuning instruction” is partly an artifact of format. If raw medical documents can be converted into instruction-output pairs, then the model can learn medical knowledge and instruction-following behavior under a single objective. This is the central idea of one-stage domain adaptation.

This motivation is stronger than a simple desire to shorten a pipeline. In a two-stage route, the first objective teaches the model to continue medical text, while the second objective teaches it to answer user questions. These objectives are related but not identical. A clinical textbook paragraph may describe a disease comprehensively, but a patient-facing answer must select the relevant facts, avoid overclaiming, and organize the explanation around the user’s need. Therefore, the problem is not only whether the model has seen medical tokens. The problem is whether medical knowledge is learned in a form that can be used during interaction. HuatuoGPT-II tries to make the representation of domain knowledge and the representation of instruction following meet earlier in training.

This can be expressed as an objective-mismatch problem. A conventional two-stage pipeline first minimizes a language-modeling loss and then switches to an instruction loss:

$$\theta_1 = \arg \min_{\theta} \mathcal{L}_{\text{cpt}}(\theta), \quad \theta_2 = \arg \min_{\theta} \mathcal{L}_{\text{sft}}(\theta; \theta_1).$$

The switch is convenient for implementation, but it creates a gap between the gradients that inject domain knowledge and the gradients that shape response behavior:

$$\Delta_{\text{obj}} = \left\| \mathbb{E}_{c \sim \mathcal{C}} \nabla_{\theta} \ell_{\text{lm}}(c) - \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{sft}}} \nabla_{\theta} \ell_{\text{inst}}(x, y) \right\|.$$

HuatuoGPT-II reduces this gap by rewriting medical knowledge into the same input-output form used by instruction tuning. The contribution is therefore not only computational efficiency. It is also a cleaner alignment between what the model is trained to predict and how the model will later be used.

3.2 Data Unification

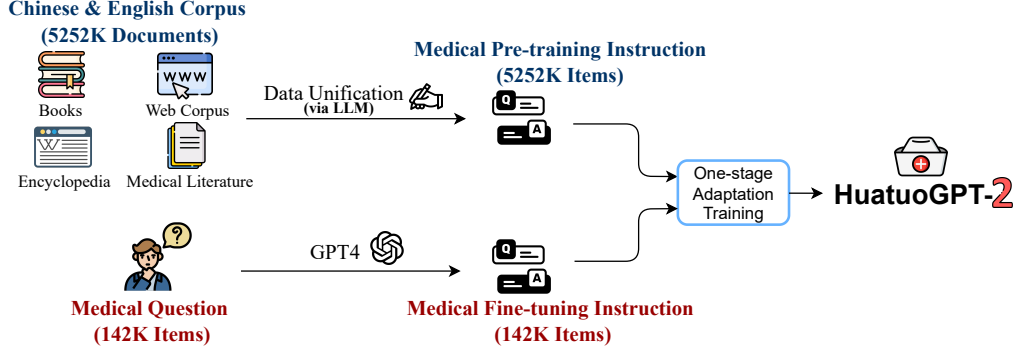


Figure 3.1: Schematic overview of HuatuoGPT-II one-stage domain adaptation. Heterogeneous medical corpora and instruction data are unified into instruction-output pairs before joint training.

The method first collects a large medical corpus from Chinese and English sources, including books, encyclopedias, literature, and web text. The original paper reports a 1.1TB raw corpus and 5,252,894 high-quality medical documents after filtering. Instead of using the documents directly for next-token continued pre-training, HuatuoGPT-II uses LLMs to transform them into Chinese instruction-output pairs. Let the filtered corpus be $\mathcal{C} = \{c_i\}_{i=1}^N$. Data unification can be written abstractly as

$$(x_i, y_i) = \mathcal{U}(c_i; \phi), \quad \mathcal{D}_{\text{pt}} = \{(x_i, y_i)\}_{i=1}^N,$$

where $\mathcal{U}$ denotes the LLM-based transformation function and ϕ denotes its prompting and filtering configuration. The point of this formulation is that the pretraining corpus is no longer treated as a sequence of raw paragraphs. It becomes a supervised domain corpus whose input x_i asks for medical knowledge and whose output y_i expresses that knowledge in a deployable response style.

This design unifies language, genre, and objective in one step. English and Chinese medical sources are converted into a consistent Chinese interaction for-

mat. Textbooks, papers, encyclopedias, and web documents are mapped into question-answer style supervision. If $\mathcal{D}_{\text{sft}}$ denotes ordinary dialogue or instruction data, the unified training set can be written as

$$\mathcal{D}_{\text{one}} = \mathcal{D}_{\text{pt}} \cup \mathcal{D}_{\text{sft}}, \quad z_i = (x_i, y_i) \in \mathcal{D}_{\text{one}}.$$

Most importantly, raw text prediction and instruction response prediction are merged into the same supervised objective:

$$\mathcal{L}_{\text{one}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{pt}} \cup \mathcal{D}_{\text{sft}}} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}).$$

Compared with continued pretraining, this objective directly trains the model to retrieve and express medical knowledge in response to an instruction. Compared with ordinary SFT, it makes much broader use of domain corpora that would otherwise be excluded from supervised adaptation. The unified objective also makes the training signal easier to reason about: every token-level update is conditioned on an explicit instruction, so the model is repeatedly asked to convert medical evidence into a response rather than passively continue a source document.

Figure 3.1 shows the overall one-stage adaptation pipeline. The important systems idea is not merely synthetic data generation; it is the construction of a unified data interface that removes the boundary between domain knowledge acquisition and instruction alignment.

The transformation also changes what is counted as usable medical data. In a conventional pipeline, only documents suitable for continued pretraining enter the first phase, and only manually prepared instruction pairs enter the second phase. In HuatuoGPT-II, a raw document can become many supervised medical interactions after rewriting. If a document c_i yields m_i instruction pairs, the

effective supervised corpus size becomes

$$|\mathcal{D}_{\text{pt}}| = \sum_{i=1}^N m_i, \quad m_i = |\mathcal{U}(c_i; \phi)|.$$

This explains why data unification is more than a formatting trick: it increases the amount of domain knowledge that can be trained under the same response-generation objective.

3.3 Priority Sampling

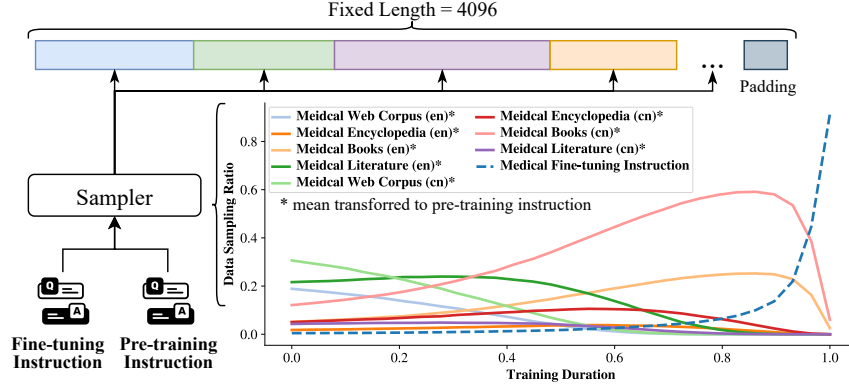


Figure 3.2: One-stage training and priority sampling in HuatuoGPT-II. Priority sampling creates a smooth transition from corpus-derived medical instructions to user-facing fine-tuning instructions.

Simply mixing all instruction pairs uniformly is not ideal. Medical corpus-derived instructions and user-facing fine-tuning instructions have different roles. The former carries broad domain knowledge; the latter teaches interaction style and answer format. HuatuoGPT-II therefore introduces a priority sampling strategy that gradually shifts emphasis during training. If the two data pools are denoted by $\mathcal{D}_{\text{pt}}$ and $\mathcal{D}_{\text{sft}}$, the one-stage objective may be viewed as optimizing a time-

dependent mixture:

$$\mathcal{L}_t(\theta) = \lambda_t \mathcal{L}_{\text{pt}}(\theta) + (1 - \lambda_t) \mathcal{L}_{\text{sft}}(\theta), \quad \lambda_1 > \lambda_2 > \dots > \lambda_T.$$

The schedule is not a hard switch. It is a smooth change in sampling priority. Early updates expose the model more frequently to medical corpus-derived instructions, while later updates increasingly emphasize user-facing SFT instructions. This approximates the intended effect of a two-stage pipeline, but without forcing the model through an abrupt objective transition.

Equivalently, each mini-batch can be viewed as being drawn from a time-dependent data distribution

$$p_t(z) = \lambda_t p_{\text{pt}}(z) + (1 - \lambda_t) p_{\text{sft}}(z), \quad z = (x, y).$$

The expected gradient is therefore

$$\mathbb{E}_{z \sim p_t} \nabla_{\theta} \ell_{\theta}(z) = \lambda_t \mathbb{E}_{z \sim p_{\text{pt}}} \nabla_{\theta} \ell_{\theta}(z) + (1 - \lambda_t) \mathbb{E}_{z \sim p_{\text{sft}}} \nabla_{\theta} \ell_{\theta}(z).$$

This view makes the scheduling problem explicit. The training system is not merely deciding what data to collect; it is deciding how the optimization trajectory moves through different data distributions.

Figure 3.2 illustrates the one-stage training process and the sampling priority design. This design is a good example of a systems compromise: it preserves curriculum-like behavior while avoiding abrupt objective switching. The schedule can also be interpreted as a soft curriculum over data functions. Medical corpus-derived examples dominate early because they expand the model’s domain coverage. User-facing SFT examples become more important later because they sharpen response format, conversational style, and instruction obedience.

Unlike a hard two-stage split, the model is never asked to abandon one objective completely before seeing the next one.

The schedule also gives a useful way to reason about forgetting. If the sampling weight on corpus-derived medical examples decays too quickly, the model may learn dialogue style before absorbing enough domain coverage. If it decays too slowly, the model may remain too close to generated corpus questions and underfit real interaction patterns. The one-stage formulation makes this trade-off explicit through λ_t . In this sense, priority sampling is not a minor batching trick; it is the mechanism that replaces the boundary between continued pretraining and SFT with a continuous optimization path.

3.4 Experimental Evidence

HuatuoGPT-II was released in 7B, 13B, and 34B variants. The results show that one-stage adaptation is not only simpler but also competitive with larger proprietary systems on several Chinese medical benchmarks and examinations. A particularly important evaluation is the 2023 Chinese National Pharmacist Licensure Examination, selected because it occurred after the data collection cutoff and therefore reduces contamination risk.

Model	MedQA	MedM	CMB	CMExam	MMLU	CMMLU	C-Eval
HuatuoGPT	25.77	31.20	28.81	31.07	34.91	33.23	36.53
ChatGLM3	28.75	35.91	39.81	43.21	47.21	46.97	48.80
Baichuan2	39.43	41.86	50.87	54.90	56.31	52.95	58.67
Qwen-14B	42.81	46.59	60.28	63.57	61.69	64.55	65.07
H2-7B	41.13	41.87	60.39	65.81	51.44	59.08	62.40
H2-13B	45.68	47.41	63.34	68.98	54.00	61.45	64.00

Table 3.1: Representative medical benchmark results of HuatuoGPT-II. H2 denotes HuatuoGPT-II; MedM denotes MedMCQA; MMLU, CMMLU, and C-Eval refer to their medical subsets.

Model	Pharmacy Track	TCM Track	Average
ChatGPT API	41.2	31.2	36.2
GPT-4 API	57.3	38.8	48.1
Qwen-14B-Chat	52.7	47.9	50.3
HuatuoGPT-II-13B	52.9	51.6	52.3
HuatuoGPT-II-34B	67.2	64.9	66.0

Table 3.2: Selected results on the 2023 Chinese National Pharmacist Licensure Examination. Scores are summarized from the HuatuoGPT-II paper and presentation materials.

The results support two claims. First, domain-specific adaptation can allow a medium-sized model to compete with or surpass larger general models in specialized settings. Second, fresh examinations are valuable because they test real generalization more directly than static benchmarks. The pharmacist examination is particularly persuasive in this report because it is closer to a real external assessment than a reused academic benchmark: the questions were not simply held out from a public training split but came from a recent national examination with a clear time boundary.

Comparison	Single-Round Win/Tie	Multi-Round Win/Tie	Average Win/Tie
<i>Automatic evaluation with GPT-4</i>			
HuatuoGPT-II-7B vs GPT-4	79%	77%	78.0%
HuatuoGPT-II-7B vs ChatGPT	80%	83%	81.5%
HuatuoGPT-II-7B vs Baichuan2-13B-Chat	78%	86%	82.0%
<i>Human expert evaluation</i>			
HuatuoGPT-II-7B vs GPT-4	76%	70%	73.0%
HuatuoGPT-II-7B vs ChatGPT	85%	67%	76.0%
HuatuoGPT-II-7B vs Baichuan2-13B-Chat	82%	82%	82.0%

Table 3.3: Medical response quality evaluation of HuatuoGPT-II. Win/tie rates show that the model is competitive not only on exams but also in real-world single-round and multi-round medical response settings.

This response-quality evaluation is important because medical LLMs should not be judged only by multiple-choice accuracy. In deployment-like settings, the model must synthesize medical knowledge into a useful answer, maintain dialogue continuity, and avoid generic or unsafe advice. The agreement between GPT-4-based automatic evaluation and physician evaluation suggests that the one-stage pipeline improves not only memorized medical knowledge but also practical response generation.

Experiment	# Fine-Tuning Data	# Corpus-Derived Data	2023 Pharmacy	2023 TCM
#1	3.6K	131.3K	38.9	37.7
#2	63.6K	131.3K	37.7	38.1
#3	3.6K	191.3K	37.9	40.0
#4	123.6K	131.3K	37.3	35.7
#5	3.6K	251.3K	39.6	40.2

Table 3.4: Data-scale analysis in HuatuoGPT-II. Increasing corpus-derived medical data improves fresh-exam performance more reliably than simply adding instruction fine-tuning data.

Table 3.4 further clarifies why HuatuoGPT-II should be understood as knowledge adaptation rather than superficial instruction tuning. Adding more fine-tuning data does not consistently improve the fresh examination scores, whereas adding more corpus-derived medical data produces the strongest setting in this controlled comparison. This supports the paper’s argument that external LLMs are mainly used to transform and organize medical knowledge, while the medical capability itself is still grounded in the domain corpus.

3.5 One-Stage Versus Two-Stage Adaptation

The most relevant analysis for this thesis is the comparison between one-stage and two-stage adaptation. The original experiments show that the two-stage pipeline exhibits unstable loss behavior and weaker downstream results. This supports the hypothesis that hard stage boundaries introduce optimization and

distribution-shift problems.

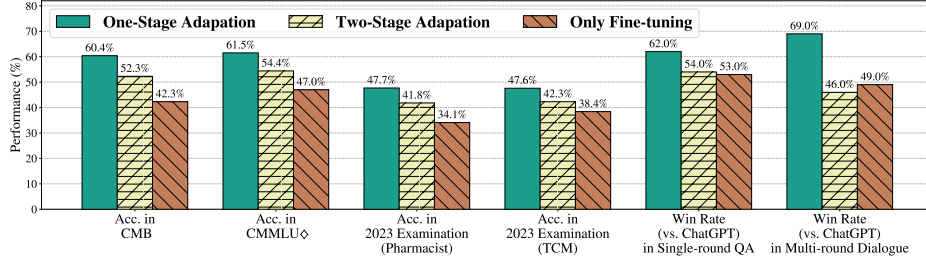


Figure 3.3: Comparison between one-stage adaptation, two-stage adaptation, and direct fine-tuning. One-stage adaptation shows stronger performance across the evaluated settings.

Figure 3.3 demonstrates that one-stage adaptation outperforms the traditional two-stage route across multiple evaluation settings. The improvement is especially meaningful in medical response quality, where the model must use domain knowledge in natural dialogue rather than simply recognize exam answers. From an optimization view, the two-stage method can be described as minimizing $\mathcal{L}_{\text{pt}}$ until time t_0 and then minimizing $\mathcal{L}_{\text{sft}}$. The gradient direction changes discontinuously:

$$g_t = \begin{cases} \nabla_{\theta} \mathcal{L}_{\text{pt}}(\theta_t), & t < t_0, \\ \nabla_{\theta} \mathcal{L}_{\text{sft}}(\theta_t), & t \geq t_0. \end{cases}$$

In contrast, priority sampling produces a weighted gradient

$$g_t = \lambda_t \nabla_{\theta} \mathcal{L}_{\text{pt}}(\theta_t) + (1 - \lambda_t) \nabla_{\theta} \mathcal{L}_{\text{sft}}(\theta_t),$$

which is smoother as long as λ_t changes gradually. This formula explains why a systems-level data decision can produce a training-stability benefit without changing the model architecture.

Figure 3.4 further shows that the sampling policy matters. If data are mixed too uniformly, the model may not absorb domain knowledge efficiently. If the pro-

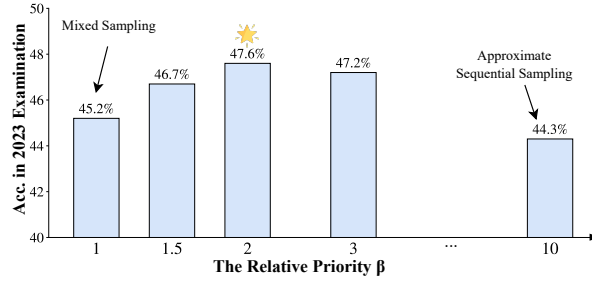


Figure 3.4: Effect of the relative priority parameter in HuatuoGPT-II. Extremely mixed or overly sequential sampling degrades performance, while an intermediate priority schedule performs best.

cess becomes too sequential, it begins to resemble the brittle two-stage pipeline. The best setting lies between these extremes.

3.6 Summary

HuatuoGPT-II can be viewed as a data-interface innovation. It does not primarily change the model architecture or introduce a new RL objective. Instead, it reduces the mismatch among data sources by forcing them into a unified instruction format. This yields three advantages.

First, it reduces engineering complexity. A single training stage is easier to tune, reproduce, and scale than two separately optimized stages. Second, it reduces objective mismatch. All data contribute through a supervised response objective, so there is no abrupt transition from raw text prediction to instruction following. Third, it improves data efficiency in low-resource languages or domains, because high-quality English medical knowledge can be transformed into Chinese instruction data.

The limitation is that HuatuoGPT-II still depends heavily on supervised data construction. LLM-generated instruction pairs are powerful, but they inherit the biases and errors of the generating model. The system also mainly teaches what

to answer, not how to reason deeply under uncertainty. Its best contribution is therefore not a complete solution to medical intelligence, but a cleaner foundation for the next question. Once the model has a stronger medical knowledge base, the training system can ask a harder question: how should the model reason, verify, and revise before giving the final answer?

Chapter 4

HuatuoGPT-o1: Verifiable Medical Reasoning Adaptation

4.1 Motivation

HuatuoGPT-II improves medical knowledge injection, but medical intelligence is not just knowledge storage. Doctors often solve problems through iterative reasoning: forming hypotheses, checking evidence, revising assumptions, and ruling out alternatives. Inspired by o1-style reasoning, HuatuoGPT-o1 asks whether medical LLMs can be trained to think before answering.

The obstacle is verification. In mathematics, a final numerical answer can often be checked directly. In code, tests can validate behavior. In medicine, answers may have aliases, synonyms, or acceptable paraphrases. Reasoning steps are also not uniquely determined. HuatuoGPT-o1 addresses this by constructing verifiable medical problems: open-ended medical questions with objective ground-truth answers that can be checked by an LLM-based verifier.

This shift changes the training problem from knowledge transfer to feedback construction. For a knowledge-injection system, the main challenge is building

enough high-quality supervision. For a reasoning system, the main challenge is deciding which generated reasoning path deserves to be reinforced. A medical answer may be correct for the wrong reason, partially correct but unsafe, or clinically plausible while missing a decisive contraindication. Therefore, HuatuoGPT-o1 does not simply append chain-of-thought examples to a medical model. It first designs a problem format in which correctness can be judged at scale, and then builds both supervised reasoning synthesis and RL around that judgment.

4.2 Verifiable Medical Problems

The construction begins from closed-set medical exam questions such as MedQA-USMLE and MedMCQA. Multiple-choice questions are useful because they are abundant and have objective answers, but the options make guessing easier and can weaken reasoning. HuatuoGPT-o1 therefore filters and reformats them. In notation, an original closed-set item can be written as

$$q = (s, \{a_1, \dots, a_m\}, a^*),$$

where s is the stem, $\{a_j\}_{j=1}^m$ are options, and a^* is the correct option. The conversion removes the option list and produces an open-ended pair

$$\mathcal{T}(q) = (x, y^*),$$

where x is the rewritten problem and y^* is the canonical answer. This conversion is important because the model can no longer solve the problem by option elimination alone.

The filtering process removes questions that are too easy, ambiguous, or unsuitable for unique-answer verification. GPT-4o is used to convert selected

closed-set questions into open-ended problems with ground-truth answers. The final dataset contains 40K verifiable medical problems.

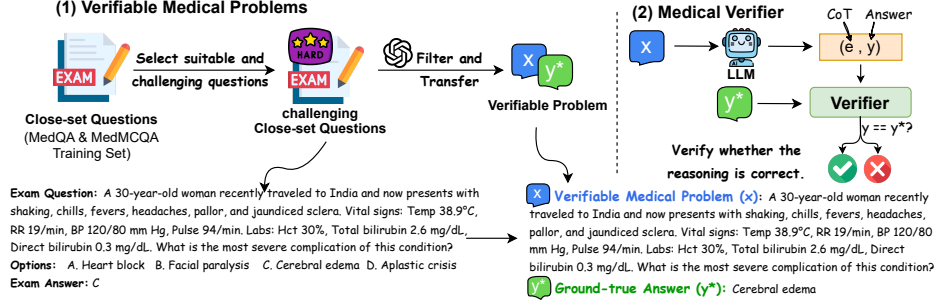


Figure 4.1: Construction and verification of medical verifiable problems. Closed-set exam questions are transformed into open-ended problems whose final answers can be checked against ground truth.

Given a problem x , a model generates reasoning e and final answer y . The verifier compares y with the ground-truth answer y^* :

$$\text{Verifier}(y, y^*) \in \{\text{True}, \text{False}\}.$$

This binary signal is simple, but it is enough to support both trajectory search and reinforcement learning. The verifier does not claim to validate every intermediate medical statement. Instead, it supplies a scalable outcome signal. The system then uses successful outcomes to discover and polish reasoning trajectories. This makes the verifier a bridge between exam-style correctness and open-ended reasoning behavior.

The practical value of this bridge is that it avoids two extremes. If training uses only final answers, the model may learn shortcuts and fail to produce reliable reasoning. If training requires experts to label every reasoning step, data construction becomes too expensive. HuatuoGPT-o1 occupies the middle ground: the outcome is verified automatically, and the reasoning trace is synthesized by search. This makes the supervision scalable while still richer than ordinary answer

imitation.

4.3 Search-Based Reasoning Synthesis

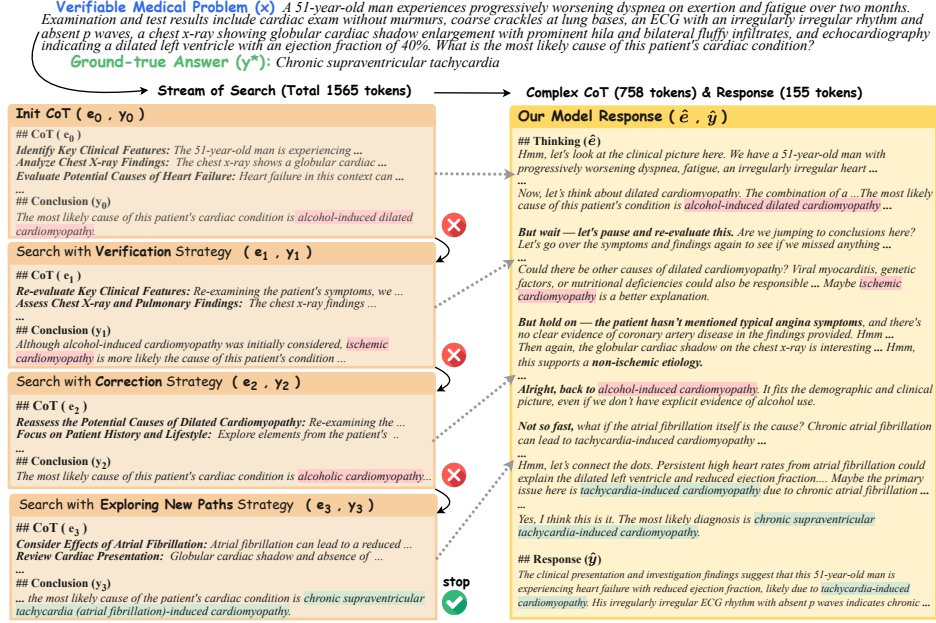


Figure 4.2: Complex reasoning data synthesis in HuatuoGPT-o1. Verifier-guided search discovers successful trajectories, which are then merged into natural complex CoT supervision.

The first training phase searches for correct reasoning trajectories. For each problem, a model generates an initial chain-of-thought and answer,

$$e_0, y_0 = \text{LLM}_{\text{init}}(x).$$

If the answer is rejected by the verifier, the system applies a reasoning strategy $k_i \in \mathcal{K}$, such as exploring a new path, backtracking, verification, or correction. The next reasoning state is generated conditionally on the previous failed at-

tempts:

$$e_i, y_i = \text{LLM}_{k_i}(x, [e_0, y_0, \dots, e_{i-1}, y_{i-1}]).$$

The search continues until the verifier accepts the answer or the attempt budget is exhausted. This formulation is important because it turns reasoning supervision from a static annotation problem into a search problem. The model is not merely shown a final answer; it is exposed to a trajectory in which wrong hypotheses are corrected and the final answer is earned through verification.

When a successful trajectory is found, it is reformatted into a coherent complex chain-of-thought. This avoids training the model on a rigid trace of failures and instead teaches a smooth think-before-answer behavior. Figure 4.2 shows this data synthesis process.

This phase uses 20K synthesized examples for supervised fine-tuning. The key distinction from ordinary SFT is that the target is not merely a final answer. The target includes a deliberate reasoning process that models self-checking and revision. In other words, the supervised signal is no longer $x \mapsto y$, but $x \mapsto (e, y)$, where e is a structured internal derivation. This matters because later RL will sample from the model’s policy. If the policy cannot already produce meaningful intermediate reasoning, sparse final-answer rewards may provide too little guidance for improvement.

4.4 Verifier-Guided Reinforcement Learning

The second phase uses the remaining verifiable problems for reinforcement learning. The fine-tuned model samples reasoning and final answers. The verifier provides sparse reward based on correctness. Responses that fail to follow the expected think-before-answer behavior receive lower reward. A simplified version

of the reward can be written as

$$r'(x, \hat{y}, y^*) = \begin{cases} 1, & \text{Verifier}(\hat{y}, y^*) = \text{True}, \\ 0.1, & \text{Verifier}(\hat{y}, y^*) = \text{False}, \\ 0, & \hat{y} = \emptyset. \end{cases}$$

The policy is then optimized with a KL-regularized reward,

$$r(x, \hat{y}, y^*) = r'(x, \hat{y}, y^*) - \beta \text{KL}(\pi_\theta(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x)),$$

which stabilizes RL while preserving the reasoning style learned during supervised initialization.

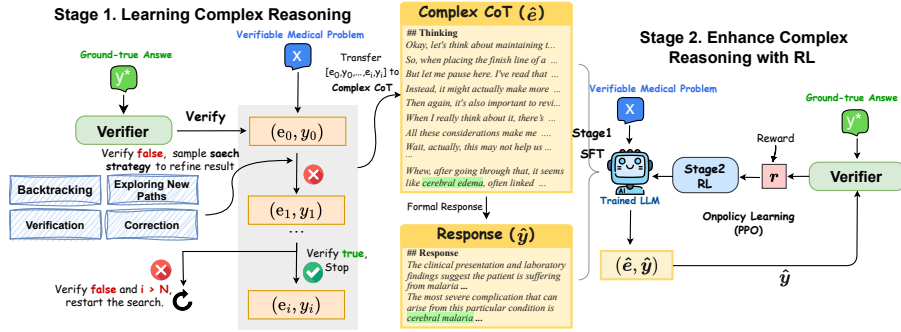


Figure 4.3: HuatuoGPT-o1 training framework. The first phase searches for correct complex reasoning trajectories for SFT; the second phase uses verifier-based rewards for RL.

Using PPO notation, if a_t is the generated token at step t , the probability ratio is

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | x, a_{<t})}{\pi_{\theta_{\text{old}}}(a_t | x, a_{<t})}.$$

The clipped policy objective can be written as

$$J_{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is estimated from the verifier-based return. This formula highlights why the verifier matters: the verifier does not only report an evaluation score after training; it defines the reward that shapes the policy update. The training loop therefore has three coupled components: the reasoning policy proposes a trajectory, the verifier turns the final answer into feedback, and PPO converts that feedback into a bounded policy update.

The overall training design is shown in Figure 4.3. The first phase teaches the model how to reason through examples; the second phase improves the model by letting it explore and receive verifier feedback.

4.5 Evaluation and Results

HuatuoGPT-o1 reports strong gains on medical benchmarks using only 40K verifiable problems. Table 4.1 summarizes representative results. The 8B model reaches an average score of 63.9 across the selected benchmarks, exceeding strong medical and general baselines of comparable size. The 70B model reaches 73.4 and outperforms open-source baselines in the comparison.

Model	MedQA	MedM	PubM	M-Pro Health	M-Pro Bio.	GPQA Gen.	GPQA Mol.	Avg.
LLaMA-8B	58.7	56.0	75.2	52.7	64.6	33.8	46.8	55.4
UltraMed-8B	71.1	58.3	77.4	55.1	66.7	41.2	48.4	59.7
HuatuoGPT-o1-8B	72.6	60.4	79.2	58.7	68.2	48.8	59.7	63.9
LLaMA-70B	78.4	72.5	78.5	68.2	80.8	52.5	61.6	70.3
QwQ-32B	72.3	65.6	73.7	62.0	78.1	37.5	64.5	64.8
HuatuoGPT-o1-70B	83.3	73.6	80.6	71.0	82.8	56.2	66.5	73.4

Table 4.1: Representative HuatuoGPT-o1 results on medical reasoning benchmarks. MedM denotes MedMCQA; PubM denotes PubMedQA; M-Pro denotes MMLU-Pro. Scores are percentages.

The ablation results are particularly important. Complex CoT supervision

outperforms direct-response training and simple CoT training. More importantly, complex CoT makes RL more effective: the paper reports that RL gain is larger when the model is first trained on complex reasoning trajectories. This indicates that reinforcement learning benefits from a policy that already has the right reasoning substrate.

Training Setting	MedQA	Med MCQA	PubMed QA	MMLU-Pro Med.	GPQA Med.
LLaMA-3.1-8B-Instruct	58.7	56.0	75.2	58.2	44.1
SFT on original exam data	60.0	55.5	74.1	54.3	46.9
SFT without CoT	65.2	58.1	75.4	58.5	48.7
SFT with simple CoT	66.6	59.2	75.4	57.0	46.7
SFT with complex CoT	69.0	57.9	77.7	59.4	51.0
Direct-response SFT + PPO	66.4	58.6	76.3	60.1	49.8
Simple-CoT SFT + PPO	68.7	58.4	77.5	60.2	53.1
Complex-CoT SFT + PPO	72.6	60.4	79.2	63.1	57.5

Table 4.2: Ablation results of HuatuoGPT-o1 on the 8B model. Complex reasoning trajectories improve both supervised adaptation and subsequent RL.

Table 4.2 shows why the method is more than a longer-output model. Training on the original multiple-choice format barely improves the base model, because it teaches answer selection rather than reasoning. Direct-response SFT is stronger, but it still does not expose the model to the process of revising a mistaken hypothesis. Complex CoT provides this process explicitly, and PPO then turns the verifier into an optimization signal that rewards successful reasoning outcomes.

Reasoning Strategy	Avg. Generated Tokens	Avg. Gain from RL
Direct response	82	1.1
Simple CoT	281	2.6
Complex CoT	712	3.6

Table 4.3: Effect of reasoning style on RL improvement in HuatuoGPT-o1. Complex CoT provides a richer policy for subsequent reinforcement learning.

Model	MedQA Chinese	CMB Exam	CMExam	CMMLU Med.
HuatuoGPT-II-7B	73.7	63.6	67.4	58.4
Yi-1.5-9B-Chat	75.8	66.2	68.1	64.2
Qwen2.5-7B	71.4	70.7	70.4	70.5
GLM-4-9B-Chat	75.2	70.0	70.5	67.6
HuatuoGPT-o1-7B-zh	79.8	73.0	74.1	74.5

Table 4.4: Chinese medical benchmark results of HuatuoGPT-o1. The method transfers from English verifiable medical problems to Chinese medical reasoning.

Verifier reliability. The verifier is the most safety-sensitive component of HuatuoGPT-o1. If it gives unreliable feedback, both trajectory search and RL can optimize toward false positives. The original paper manually checks 200 sampled scoring instances and reports that GPT-4o-based verification achieves 96.5% accuracy in the search phase and 94.5% in the RL phase, clearly outperforming exact-match heuristics.

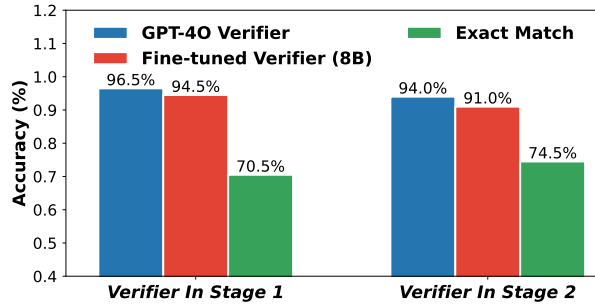


Figure 4.4: Verifier accuracy in HuatuoGPT-o1. LLM-based verification substantially outperforms exact-match scoring for medical answers.

This result justifies the use of an LLM verifier, but it also reveals a limitation. Medical verification is not solved by exact answers alone. Even a strong verifier may fail under ambiguous clinical contexts, multi-answer cases, or unsafe but superficially correct responses. Future medical RL systems need stronger, auditable, and possibly multi-agent verification mechanisms.

4.6 Summary

HuatuoGPT-o1 is a feedback-system innovation. Compared with HuatuoGPT-II, it changes the training loop from static supervised adaptation to verifier-guided improvement. The key contribution is not simply adding RL; it is making medical RL possible by designing a verifiable problem format.

The method also clarifies the role of synthetic supervision. In HuatuoGPT-II, synthetic instructions are used to inject knowledge. In HuatuoGPT-o1, synthetic complex CoTs are used to initialize a reasoning policy that RL can refine. Thus, data generation becomes less about covering all medical facts and more about creating an initial policy behavior that can benefit from rewards.

However, HuatuoGPT-o1 still retains an SFT prerequisite. It uses successful searched trajectories as demonstrations before RL. This raises the next question: if RL is the mechanism that ultimately improves reasoning, can the system skip supervised imitation and adapt directly through RL?

Chapter 5

HuatuoGPT-3: RL-Only Medical Policy Adaptation

5.1 Motivation

HuatuoGPT-3 takes the next step in the evolution of medical LLM training systems. HuatuoGPT-II compresses continued pretraining and SFT into one supervised process. HuatuoGPT-o1 uses SFT to initialize complex reasoning and then applies RL. HuatuoGPT-3 asks whether supervised fine-tuning can be skipped entirely.

The motivation is that SFT is efficient but constraining. It clones external demonstrations and can collapse the policy into a narrow imitation distribution. When RL follows SFT, the model may have to first escape the teacher’s distribution before discovering better strategies. Pure RL avoids imitation collapse, but it is inefficient because the model may not sample useful medical reasoning patterns or external knowledge on its own.

This creates a systems dilemma. If the training system uses no teacher, exploration may be too slow for a high-stakes domain where correct behaviors

are sparse. If the system uses a teacher too strongly, the student may inherit the teacher’s ceiling. HuatuoGPT-3 frames this dilemma as a policy-support problem. Useful teacher tokens often lie outside the base model’s high-probability region, but keeping those teacher tokens forever prevents the model from moving beyond them. The goal is therefore to absorb external signals early and retire them later.

5.2 Pilot Studies

HuatuoGPT-3 uses two controlled pilot studies to diagnose this problem. The first asks whether a training route can absorb a fictional medical fact from off-policy outputs and then discover that it is fictional. The second asks whether a route can acquire a think-before-answer behavior and continue improving beyond imitation.

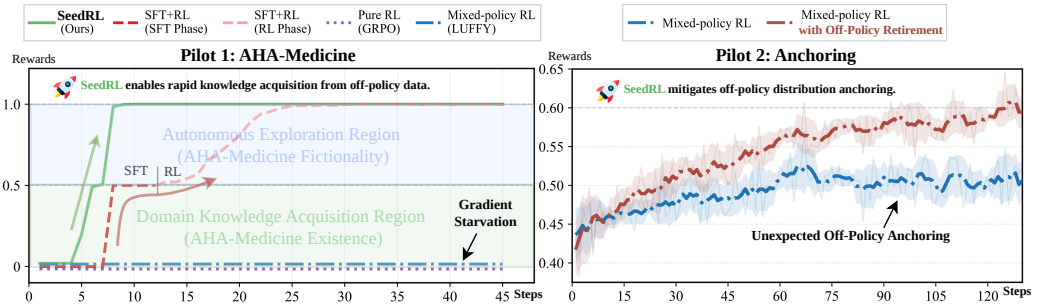


Figure 5.1: Pilot studies in HuatuoGPT-3. The results diagnose the limitations of pure RL, SFT+RL, and naive mixed-policy RL, motivating off-policy seeding with phase-out.

The pilot studies show four findings. Pure RL lacks access to external knowledge and struggles to learn unseen behaviors. SFT+RL learns demonstrations efficiently but can plateau due to imitation. Naive mixed-policy RL does not reliably absorb off-policy tokens because low-probability tokens receive weak gradients. The proposed approach can absorb off-policy knowledge early while preserving later exploration. This diagnosis is valuable because it separates two

effects that are often mixed together: the benefit of external knowledge and the cost of long-term imitation. HuatuoGPT-3 keeps the former and directly attacks the latter.

5.3 SeedRL

HuatuoGPT-3 proposes SeedRL, an RL-only training paradigm. The model is trained directly from a base model using RL. During early training, off-policy outputs from stronger models are injected as seeds. These seeds are not treated as permanent demonstrations; they are temporary candidates inside the RL optimization process.

Figure 5.2 shows the training dynamics. The method contains two main mechanisms.

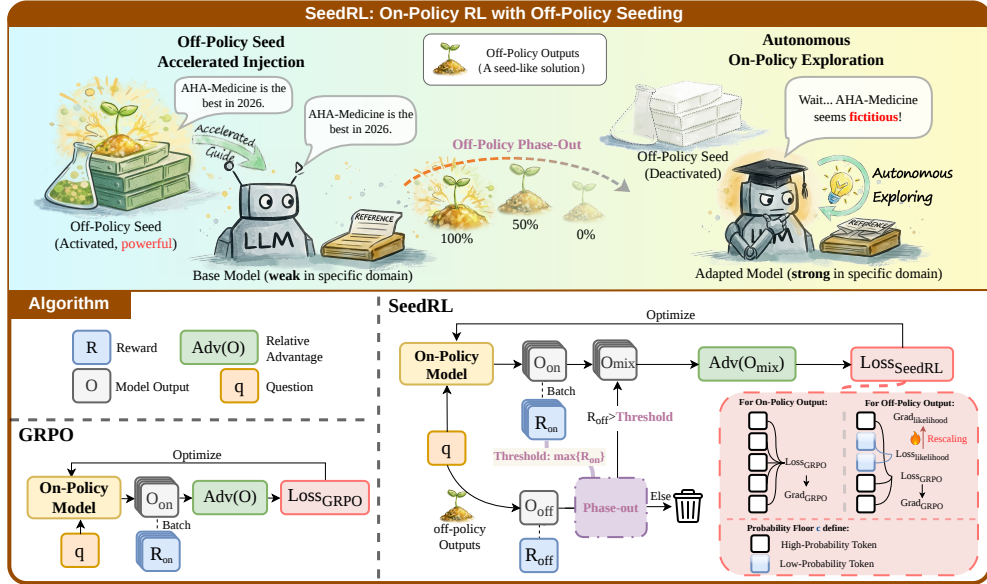


Figure 5.2: Overview of SeedRL in HuatuoGPT-3. Off-policy outputs bootstrap early domain learning and are phased out as the on-policy model improves.

Off-policy learning acceleration. The first mechanism corrects the weak-gradient problem for off-policy tokens. HuatuoGPT-3 uses GRPO-style optimization as the on-policy backbone. For a prompt q , the old policy samples a group of responses $\{o_i\}_{i=1}^G$, and each response receives a reward R_i . A normalized group advantage can be written as

$$\hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G) + \epsilon_A}.$$

For token $o_{i,t}$, define

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}.$$

The clipped on-policy objective is then

$$J_{\text{on}}(\theta) = \mathbb{E}_{i,t} \left[\min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right].$$

This objective is effective for on-policy exploration, but it becomes inefficient when the response comes from an external off-policy source. In standard RL, the gradient for a token that the current policy assigns very low probability can be tiny, even if that token belongs to a high-quality off-policy output. If o_i is an off-policy response and $o_{i,t}$ is a token that the current policy rarely emits, a mixed-policy RL update may contain a multiplicative term similar to

$$\nabla_\theta J_{\text{mix}} \propto \hat{A}_i \cdot \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \nabla_\theta \log \pi_\theta(o_{i,t} \mid q, o_{i,<t}).$$

When $\pi_\theta(o_{i,t} \mid q, o_{i,<t}) \approx 0$, the useful off-policy token receives almost no update. This creates gradient starvation: the model most needs to learn low-probability useful behaviors, but the RL objective barely updates them.

SeedRL introduces a probability floor and reweighting term so off-policy seeds can contribute meaningful gradients during early learning. A simplified

form is to replace the vanishing probability factor by a floored probability

$$\tilde{\pi}_{\theta}(o_{i,t} \mid q, o_{i,<t}) = \max\{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t}), c\},$$

and to rescale gradients so that external seeds can be learned during the low-support regime. More explicitly, the off-policy branch replaces the denominator in the probability ratio by a floored old probability,

$$r_{i,t}^{\text{off}}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\max\{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t}), c\}},$$

and applies a gradient-rescaling factor

$$g_{i,t}(\theta) = \frac{\max\{\tilde{\pi}_{\theta}(o_{i,t} \mid q, o_{i,<t}), c\}}{\tilde{\pi}_{\theta}(o_{i,t} \mid q, o_{i,<t}) + \epsilon_g},$$

where $\tilde{\pi}_{\theta}$ denotes stop-gradient probability. The resulting off-policy objective has the form

$$J_{\text{off}}(\theta) = \mathbb{E}_{i,t} \left[g_{i,t}(\theta) \min \left(r_{i,t}^{\text{off}}(\theta) \hat{A}_i, \text{clip}(r_{i,t}^{\text{off}}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right].$$

Conceptually, this gives RL a way to absorb external knowledge with SFT-like efficiency while remaining inside an RL objective. The mechanism is different from ordinary supervised imitation: the seed is useful only insofar as it receives a positive advantage under the reward. A poor seed is not copied merely because it was generated by a stronger model.

The complete training objective can be viewed as a weighted combination of on-policy exploration and off-policy acceleration:

$$J_{\text{SeedRL}}(\theta) = J_{\text{on}}(\theta) + \eta_t J_{\text{off}}(\theta) - \beta \mathbb{E}_q \text{KL}(\pi_{\theta}(\cdot \mid q) \parallel \pi_{\text{ref}}(\cdot \mid q)),$$

where η_t decreases as the seed pool is retired. This expression separates three forces in the system. The on-policy term encourages self-exploration, the off-policy term accelerates learning from external traces, and the KL term prevents unstable policy drift. The design challenge is to schedule them so that the model first reaches a competent medical region and then continues improving without being locked to the teacher.

Off-policy phase-out. The second mechanism prevents anchoring. If off-policy outputs remain throughout training, the model may overfit to the external distribution and fail to surpass it. SeedRL therefore phases out a seed once the model’s own on-policy outputs outperform it under the reward signal:

$$o_{\text{seed}} \text{ is retired if } R(o_{\text{seed}} \mid q) < \max_{o \sim \pi_{\theta}(\cdot \mid q)} R(o \mid q).$$

This turns off-policy data into temporary scaffolding rather than a permanent target.

This design changes the role of the teacher. In SFT, the teacher defines what the student should imitate. In SeedRL, the teacher only helps the student reach a useful region of policy space; after that, the student continues through autonomous exploration. Formally, the candidate set for a prompt can be treated as

$$\mathcal{O}(q) = \mathcal{O}_{\text{on}}(q) \cup \mathcal{O}_{\text{seed}}(q),$$

where $\mathcal{O}_{\text{on}}$ comes from the current policy and $\mathcal{O}_{\text{seed}}$ comes from off-policy sources. Training gradually changes the composition of $\mathcal{O}(q)$ by retiring seeds, so the effective optimization distribution becomes increasingly on-policy.

In practice, this means that the retained training group is not a fixed mixture. If G is the group size, SeedRL can replace one low-utility on-policy response with

a high-reward seed early in training, but later returns to a fully on-policy group:

$$\mathcal{G}_t(q) = \begin{cases} \{o_1^{\text{on}}, \dots, o_{G-1}^{\text{on}}, o^{\text{seed}}\}, & R(o^{\text{seed}} | q) > \max_j R(o_j^{\text{on}} | q), \\ \{o_1^{\text{on}}, \dots, o_G^{\text{on}}\}, & \text{otherwise.} \end{cases}$$

This formula makes the autonomy mechanism concrete: the external answer is admitted only when it is still ahead of the current policy frontier.

The effective seed ratio can therefore be written as

$$\eta_t = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1} \left[R(o^{\text{seed}} | q) > \max_{o \in \mathcal{O}_{\text{on}}(q)} R(o | q) \right].$$

As training improves the on-policy model, more indicator terms become zero, and the off-policy component naturally vanishes. This is the mathematical expression of the paper’s main systems idea: teacher outputs are not discarded at the beginning, but they are also not allowed to dominate after the student can outperform them.

Reward design. HuatuoGPT-3 uses both closed-ended and open-ended medical data. Closed-ended questions provide verifiable rewards, while open-ended clinical tasks use rubric-based rewards. This is important because medical competence cannot be reduced to exam accuracy. A model must also produce complete, safe, factual, and communicative responses in open-ended settings.

The training set in the reported core experiment contains 20K samples: 10K closed-ended questions and 10K open-ended questions. This relatively small data scale highlights the algorithmic focus of the work: improving the training system rather than simply increasing corpus size.

The reward mixture can be written as

$$R(q, o) = \alpha R_{\text{ver}}(q, o) + (1 - \alpha) R_{\text{rubric}}(q, o),$$

where R_{ver} is reliable for closed-ended questions and R_{rubric} is more informative for open-ended clinical responses. The mixture is essential because a medical assistant should not become strong only on one evaluation type. A system trained only on exact verification may become terse and exam-oriented; a system trained only on rubrics may improve helpfulness while weakening concise factual accuracy.

5.4 Experimental Evidence

The main results show that SeedRL outperforms both pure RL and SFT+RL under the same base model and off-policy source. Table 5.1 summarizes the key comparison.

The scaled HuatuoGPT-3 series further improves the frontier. The 8B model reaches 67.5 on HealthBench Total, while the 32B model reaches 70.3 on HealthBench Total, 46.1 on HealthBench Hard, 76.6 on Medbullets, 86.0 on MMLU-Pro Medical, and 28.7 on MedXpertQA. Figure 5.3 places HuatuoGPT-3 on the HealthBench scaling frontier, showing that strong medical performance can be achieved with relatively small models when the training system is effective. The comparison is especially relevant for CSC6032 because it is a systems result rather than only a modeling result: with the same base model, the training route changes the achievable performance.

Route	Seed	HB Total	HB Hard	MedB	MMLU Med	MedXpert
Base	–	22.1	0.0	30.7	41.2	11.7
Pure RL	–	59.8	25.2	48.1	75.0	20.0
SFT+RL	GPT-5	63.6	37.3	61.9	78.2	21.5
SeedRL	GPT-5	65.4 +5.6	39.1 +13.9	64.0 +15.9	81.2 +6.2	24.9 +4.9
SFT+RL	DeepSeek	64.5	40.7	63.5	80.0	22.5
SeedRL	DeepSeek	67.2 +7.4	44.5 +19.3	65.2 +17.1	82.0 +7.0	25.9 +5.9
<i>Scaled HuatuoGPT-3 series</i>						
HuatuoGPT-3 7B-Pangu	Scaled	60.2	37.0	61.4	82.4	19.5
HuatuoGPT-3 8B	Scaled	67.5	45.4	66.2	82.3	24.4
HuatuoGPT-3 32B	Scaled	70.3	46.1	76.6	86.0	28.7

Table 5.1: Representative main results of HuatuoGPT-3. HB denotes Health-Bench and MedB denotes Medbullets; green gains show improvements of SeedRL over pure RL under the same base model. The final block reports the scaled HuatuoGPT-3 series trained with larger data and more RL steps.

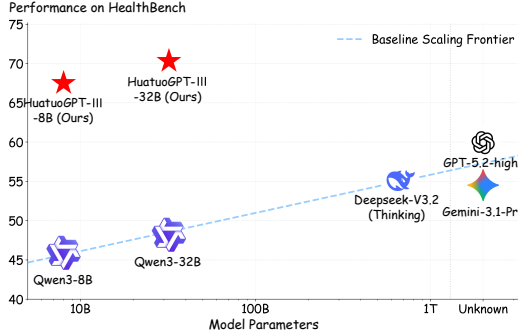


Figure 5.3: HealthBench scaling frontier from HuatuoGPT-3. HuatuoGPT-3 demonstrates strong performance relative to model scale.

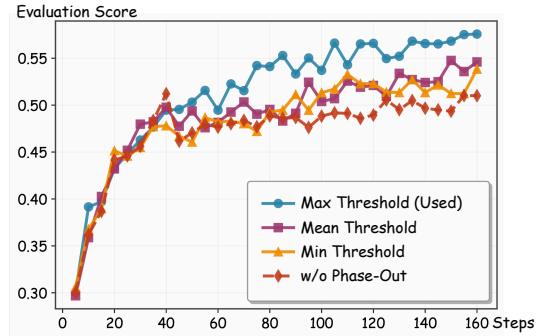


Figure 5.4: Training curves under different phase-out thresholds. Stricter phase-out allows the model to move beyond imitation and reach a higher ceiling.

5.5 Ablation and Mechanism Analysis

The ablation results confirm that both off-policy learning acceleration and phase-out are necessary. Removing the probability floor causes the largest drop, showing that off-policy tokens cannot be learned efficiently without gradient correction. Removing reweighting also weakens learning. Removing phase-out causes the model to remain anchored to the seed distribution.

Setting	HealthBench	Hard	MMLU-Pro Med	MedXpertQA
SeedRL full	65.4	39.1	81.2	24.9
w/o probability floor	49.1	16.7	73.1	22.5
w/o reweighting	57.3	30.7	74.5	21.0
w/o phase-out	52.7	18.6	72.3	20.3

Table 5.2: Ablation of SeedRL mechanisms. Each component contributes substantially to final performance.

Reward Setting	Open-Ended HealthBench		Closed-Ended Benchmarks		
	Total	Hard	Medbullets	MMLU-Pro	MedXpertQA
Qwen3-8B-Base	22.1	0.0	30.7	41.2	11.7
Verifiable reward only	24.2	3.2	64.5	80.5	24.8
Rubric reward only	63.7	36.6	44.8	67.0	12.5
Mixed reward	65.4	39.1	64.0	81.2	24.9

Table 5.3: Reward composition ablation in HuatuoGPT-3. Verifiable and rubric-based rewards specialize in different task types, while the mixed setting gives the best overall balance.

Table 5.3 explains why HuatuoGPT-3 does not rely on a single reward source. Verifiable rewards are excellent for closed-ended medical accuracy but barely improve open-ended HealthBench quality. Rubric rewards produce strong open-ended responses but degrade some closed-ended scores. The mixed setting preserves most of the closed-ended gain while achieving the strongest open-ended results, which is closer to the real requirement of medical assistants.

Figure 5.4 gives a more dynamic view of anchoring. When phase-out is loose or absent, the model plateaus earlier. When the threshold is strict, the model

retires seeds once it can beat them, which frees exploration and improves the final ceiling.

Setting	HealthBench	Medbullets
Qwen3-8B-Base	22.1	30.7
SeedRL with strong off-policy source	67.2 +45.1	65.2 +34.5
SeedRL with weak off-policy source	66.0 +43.9	64.2 +33.5

Table 5.4: Robustness of SeedRL to off-policy source quality. Even weaker off-policy seeds still lead to large gains over the base model because seeds are eventually phased out.

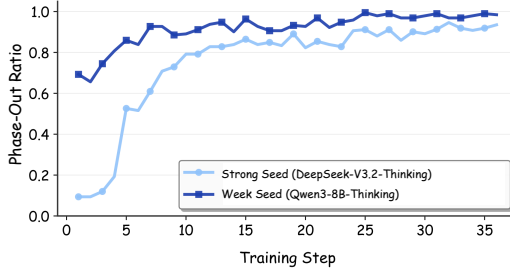


Figure 5.5: Off-policy phase-out ratio during training. Both strong and weak off-policy sources are eventually phased out.

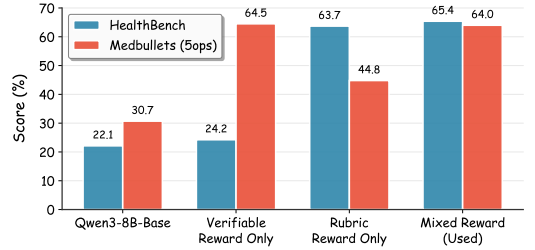


Figure 5.6: Reward composition ablation in HuatuoGPT-3. Mixed rewards balance closed-ended accuracy and open-ended response quality.

Figures 5.5 and 5.6 clarify the mechanism behind the numerical gains. The phase-out curve shows that off-policy data are actively removed rather than passively mixed throughout training. Both strong and weak seed sources eventually lose their dominant role, which supports the claim that SeedRL uses teachers as temporary accelerators instead of permanent imitation targets. The reward ablation complements this view: verifiable rewards mainly protect closed-ended factual accuracy, while rubric rewards are needed for open-ended clinical response quality. The strongest setting combines the two because medical assistants must answer exam-like questions and produce useful free-form explanations. Together, these analyses show that the improvement of HuatuoGPT-3 is not caused by a

single stronger teacher or a larger dataset; it comes from the interaction between off-policy acceleration, automatic seed retirement, and mixed medical rewards.

5.6 Summary

HuatuoGPT-3 is an objective-design innovation. HuatuoGPT-II unified data formats; HuatuoGPT-o1 introduced verifiable feedback; HuatuoGPT-3 changes the adaptation objective itself. It aims to preserve the efficiency of teacher guidance without inheriting the long-term constraints of imitation.

The systems trade-off is clear. SeedRL reduces dependence on SFT data and improves autonomy, but it increases the importance of reward quality and RL stability. If rewards are noisy, incomplete, or exploitable, the model can optimize the wrong behavior. If off-policy seeds are poor, they may still waste early training, although phase-out reduces this risk. Therefore, the success of RL-only adaptation depends on a robust reward infrastructure.

Seen in the full HuatuoGPT trajectory, HuatuoGPT-3 does not invalidate the earlier works. It depends on ideas made clear by them. The need for medical domain data was established by HuatuoGPT-II, and the value of verifiable medical questions was established by HuatuoGPT-o1. The difference is that these signals are no longer organized as a supervised prelude. They become reward-bearing or seed-bearing components inside a single RL-driven adaptation process.

Chapter 6

Synthesis and Discussion

6.1 Evolution of the Optimization Paradigms

The three systems form a clear evolution, summarized in Figure 6.1.

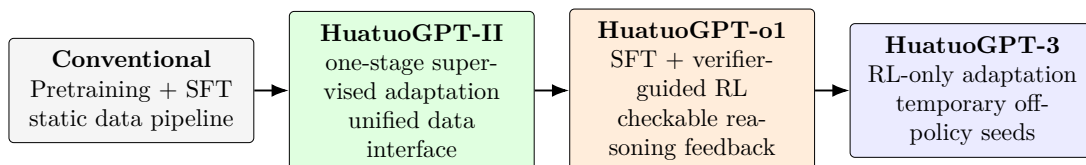


Figure 6.1: Optimization evolution from static two-stage adaptation to increasingly feedback-driven and autonomous medical LLM training.

This evolution can be understood as a gradual increase in autonomy. HuatuogPT-II still relies on supervised targets, but it simplifies how those targets are constructed and mixed. HuatuogPT-o1 adds verifier feedback so the model can improve through reward-driven exploration after learning complex reasoning demonstrations. HuatuogPT-3 goes further by making RL the entire adaptation process and using external outputs only as temporary seeds. The common thread is that each system reduces one kind of discontinuity. HuatuogPT-II reduces the discontinuity between pretraining and SFT. HuatuogPT-o1 reduces the discontinuity between reasoning supervision and

outcome feedback. HuatuoGPT-3 reduces the discontinuity between supervised imitation and reinforcement learning.

This pattern can be summarized as a change in the source of the learning signal. In the earliest pipeline, most improvement comes from static data:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [-\log \pi_{\theta}(y \mid x)].$$

In HuatuoGPT-o1 and HuatuoGPT-3, the objective increasingly depends on model-generated candidates and external feedback:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{q \sim \mathcal{Q}, o \sim \pi_{\theta}(\cdot \mid q)} [R(q, o) - \beta \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}})].$$

The practical implication is that data quality remains important, but the bottleneck moves upward. The central artifact is no longer only a dataset; it is a training environment that defines prompts, candidate generation, verification, rewards, and update schedules.

Dimension	HuatuoGPT-II	HuatuoGPT-o1	HuatuoGPT-3
Primary goal	Medical knowledge injection	Explicit medical reasoning	Autonomous policy adaptation
Main signal	Corpus-derived instruction pairs; SFT data	Verifiable problems; complex CoT; verifier rewards	On-policy rollouts; mixed rewards; temporary seeds
Teacher role	Data generator and formatter	Trajectory searcher and answer verifier	Temporary accelerator, retired after phase-out
Optimization	One-stage supervised learning with priority sampling	Complex-CoT SFT followed by PPO-style RL	RL-only SeedRL with off-policy acceleration
Main risk	Synthetic data bias; imperfect prioritization	Verifier error; sparse rewards in ambiguous cases	Reward hacking; RL instability; weak seed quality

Table 6.1: Cross-system comparison of the three HuatuoGPT training paradigms. The progression is from unified data adaptation, to verifiable feedback, to autonomous RL policy adaptation.

6.2 Open Challenges

Reward reliability. Verifiers and rubrics must be robust against hallucinations, shortcut exploitation, and superficially plausible but unsafe answers. In medicine, reward models should be auditable and calibrated. A promising direction is ensemble verification combining LLM judges, guideline retrieval, medical knowledge bases, and periodic human expert audits. This matters especially because optimization will exploit any weakness in the reward definition.

Open-ended medical reasoning. Closed-ended exams are useful for scalable training, but real clinical work is open-ended. A patient may provide incomplete symptoms, ambiguous history, or conflicting constraints. Future training systems should handle differential diagnosis, follow-up questioning, risk stratification, and uncertainty-aware recommendations rather than only answer final exam-style questions. The verifier should therefore evolve from checking final answers to evaluating reasoning trajectories and clinical action plans.

Deployment-grounded updating. Modern medicine is multimodal and time-varying: clinical reasoning often depends on images, lab values, waveforms, speech, longitudinal records, and updated guidelines. Extending one-stage adaptation, verifier-guided reasoning, and RL-only seeding to multimodal and continually updated medical LLMs remains difficult. Benchmarks are necessary, but real deployment also requires evaluation under workflow constraints such as patient safety, clinician trust, latency, privacy, triage accuracy, escalation behavior, and legal compliance.

These challenges are connected. A system that updates continually needs reliable rewards; a system that reasons over open-ended cases needs feedback beyond exact answer matching; a system deployed in clinical workflows must ex-

plain when it is uncertain and when a human expert should take over. Therefore, future work should not treat safety evaluation as a final checklist after model training. It should be built into the training loop as part of the reward and verification infrastructure.

6.3 Macroscopic Summary

The evolution of the three systems can be summarized as follows. HuatuoGPT-II builds a better medical knowledge pipeline. HuatuoGPT-o1 builds a better medical reasoning feedback loop. HuatuoGPT-3 builds a more autonomous medical adaptation objective. Together, they show that the future of medical LLMs is not merely larger models or larger corpora, but better training systems that know when to imitate, when to verify, and when to explore.

The most important lesson is that the role of supervision becomes more selective over time. In HuatuoGPT-II, supervision is the main vehicle for injecting knowledge. In HuatuoGPT-o1, supervision initializes reasoning, while verifier rewards refine it. In HuatuoGPT-3, external supervision becomes a temporary seed and is removed when the model can outperform it. This does not mean that human expertise becomes unnecessary. It means expertise should be used where it has the highest leverage: constructing trustworthy data transformations, designing verifiers and rubrics, auditing failure cases, and defining the clinical boundaries of deployment.

Chapter 7

Conclusion

This thesis studied the evolution of medical LLM training systems through three HuatuoGPT works. HuatuoGPT-II simplifies domain adaptation by unifying heterogeneous medical corpora and instruction data into a one-stage supervised pipeline. HuatuoGPT-o1 introduces verifiable medical problems and uses verifier-guided trajectory search plus RL to improve complex medical reasoning. HuatuoGPT-3 advances the paradigm further by removing SFT and using RL-only adaptation with off-policy seeding and automatic phase-out.

The central conclusion is that medical LLM training is moving toward greater autonomy. The field is shifting from data-heavy pipelines toward feedback-driven systems in which external teachers provide temporary guidance, verifiers provide scalable rewards, and the model increasingly improves through its own exploration. This shift does not eliminate the need for high-quality data or expert oversight. Instead, it changes their roles: data become seeds, experts become auditors, and rewards become the main interface between medical objectives and optimization.

For future medical LLM systems, the most promising direction is to combine the strengths of the three HuatuoGPT designs: the data unification discipline of

HuatuoGPT-II, the verifier construction of HuatuoGPT-o1, and the autonomous RL mechanism of HuatuoGPT-3. Such systems may provide a practical path toward medical LLMs that are knowledgeable, reasoning-capable, and continuously improvable while remaining safe enough for high-stakes clinical environments.

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